

JACOB COOK

Cybersecurity Engineer II | Cloud Security · Azure AD · WAF/API Concepts · Threat Modeling · HITRUST/CIS · Python · IaC

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Cloud-leaning security engineer with hands-on Azure AD, AWS lab work, and SIEM detection engineering. Built Wazuh from scratch, authored Sigma-format detection-as-code, and shipped FastAPI-based SOAR orchestration with webhook ingestion and pytest-backed validation. Comfortable with threat modeling, change-managed deployments, and CIS/HITRUST-aligned controls. CompTIA stack (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress; six years of monitoring, RCA, and incident coordination.

TECHNICAL SKILLS

Cloud & Infrastructure Security: Azure AD, AWS labs, Cloud-Native Policy Tuning, Web Application Firewall (WAF) Concepts, API Management, Infrastructure-as-Code (Terraform familiar), CI/CD Security
Detection & Incident Response: Threat Modeling, Incident Triage & Documentation, Containment, Root Cause Analysis, SLA-Driven Response, Change Management Coordination, Detection Engineering
Compliance & Frameworks: HITRUST & CIS Familiarity, NIST 800-53/61, MITRE ATT&CK, Vulnerability Management, Risk-Based Prioritization, Audit Evidence Preparation
Languages & Platforms: Python, Bash, PowerShell, SQL, FastAPI, REST APIs, Git; SIEM (Wazuh hands-on), Linux, Windows, Wireshark

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code 2026

- Authored 6 Sigma-format detection rules for Intune MDM abuse mapped to MITRE ATT&CK, with enrichment logic that eliminated 100% of false positives in a 100-event simulation (80 benign suppressed, 20 malicious caught) using KV store-style admin baseline lookups.
- Documented 3 benign false-positive scenarios per rule (18 total), covering VPN logins, scheduled maintenance, and PIM activations so analysts can triage confidently without escalating known-good activity.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py), reproducible against synthetic data, not a live tenant.

Blue Team SOC Monitoring Lab: Wazuh SIEM 2026

- Built a full Wazuh SIEM stack from scratch and validated 12 detection rule patterns against 242 simulated auth-style events, separating true-positive threats from benign noise with 100% precision after tuning.
- Reduced false-positive alert volume by 100% by tuning out all 79 benign-only matches (cron jobs, service accounts, scheduled deployments) without dropping a single true positive.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py). All metrics reproducible.

SOAR-lite: Incident Response Orchestrator 2026

- Eliminated 4 manual triage steps (classification, IOC enrichment, notification, ticket creation) by building automated SIEM-triggered playbooks, reducing triage time by over 60% in lab testing.
- Built end-to-end alert automation: webhook ingestion from Wazuh, automatic playbook routing by event type, IP blocking, Slack notification, and incident documentation with zero analyst intervention required.
- Delivered a working handoff path from initial alert handling to documented, Tier 2-ready response that closes the gap between raw detection and actionable cases.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py). Tests run against the real FastAPI app, not mocked endpoints.

Phishing Analysis Lab 2026

- Triaged 5 real-world phishing samples end-to-end: header inspection, SPF/DKIM/DMARC validation, URL defanging, VirusTotal enrichment, IOC extraction, and attacker infrastructure mapping, achieving 100% accurate threat classification against known IOC ground truth.
- Delivered production-style triage reports with documented IOCs, confidence assessments, and escalation decisions that mirror real security operations output.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6) 2024

- Secured a production AI platform for corporate stakeholders by engineering RBAC and Azure AD SSO, enforcing data privacy requirements that enabled executive-level adoption.
- Shipped an enterprise-grade platform on deadline as part of a six-person team under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response 2019 - 2025

- Maintained 99.5% system uptime across automated sortation and conveyor infrastructure by monitoring sensors, alerts, and operational dashboards across rotating shifts, the same continuous monitoring discipline that defines round-the-clock SOC work.
- Resolved 200+ system anomalies over six years through structured root cause analysis, producing compliance-tracked documentation that reduced repeat incidents by an estimated 30% and supported audit readiness.
- Reduced mean time to recovery by an estimated 20% by coordinating cross-team incident response during outages, including escalation, triage prioritization, and post-incident reporting that mirrors security incident response workflows.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress) Expected Oct 2026 · GPA 3.50
CSUS | BS, Computer Science Jan 2025 · GPA 3.50
CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)